

Rafael A. Dutra

AI Developer — Machine Learning Engineering

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PROFESSIONAL SUMMARY

AI Developer with hands-on experience in LLMs, multi-agent systems, RAG, and Machine Learning. At the Porto Digital/Globo Technology Residency, I developed video processing pipelines and ML models for automatic anomaly detection in live broadcasts. Experience in Data Analysis, Power BI, and the full ML cycle: data collection, modeling, containerization, and monitoring. C2 English (EFSET).

TECHNICAL SKILLS

- **AI & Machine Learning:** LLMs (GPT-4o), RAG, Autonomous Agents, NLP, Prompt Engineering, Computer Vision, InsightFace, Scikit-learn, TensorFlow, PyTorch, XGBoost, SHAP
- **Data & Analytics:** ETL, Data Pipelines, Data Cleaning, Business Insights, Power BI, SQL, Pandas, NumPy, Streamlit, Jupyter, Excel
- **Languages & Frameworks:** Python, JavaScript, FastAPI, React, Angular, Biopython
- **Infrastructure & Data:** PostgreSQL, MongoDB, Redis, Docker, Docker Compose, Git/GitHub, Linux, REST APIs

PROFESSIONAL EXPERIENCE

AI Developer — Porto Digital / Globo Technology Residency — Recife, PE (Academic-Business Partnership) Feb/2024 — Present

- Developed Horus AI: 3 ML models (video 97.6%, audio 90.9%, lipsync 100%) for automatic anomaly detection in live broadcasts, replacing manual monitoring with real-time detection.
- Implemented a video collection and processing pipeline (MP4, WMV, HLS/SRT) and integrated the Angular frontend, FastAPI backend, and ML models in a Docker environment.

AI Developer — Identity Service Project — Volunteer 2024

- Developed a facial recognition microservice with InsightFace/ArcFace (512-d ArcFace embedding, cosine similarity) exposed via FastAPI, delivered in a staging environment with Docker and documented REST endpoints.

HIGHLIGHT PROJECTS

Horus AI — Anomaly Detection in Streaming • Python · FastAPI · TensorFlow · OpenCV · Angular · WebSocket | Feb/2024 — Present

github.com/HaelI39/horus_ai

- Hybrid OpenCV + ML strategy with temporal voting; automatic clip generation as evidence, WebSocket alerts, dashboard with KPIs, and PDF report export.

Retail Intelligence — Data, Logistics & Strategy • Python · Scikit-learn · Pandas · XGBoost · Power BI | 2024

github.com/HaelI39/inteligencia-varejo-dinamica-consumo

- Analysis of Brazilian e-commerce (Olist + IBGE dataset): delay classification model (96.7% accuracy) and satisfaction regression, identifying Lead Time as the main driver vs. inflation.
- Insight: Pernambuco logistics with +116% Lead Time vs. SP; delays reduce satisfaction from 4.29★ to 2.27★ (-47%); customer retention ~0–1% after 1 month.

MedNode — Multi-Agent Clinical Support • Python · FastAPI · PostgreSQL · React · Docker · GPT-4o-mini | Feb/2024 — ongoing

- Micro-SaaS with 3 orchestrated autonomous agents: RAG pipeline with fallback across 4 scientific APIs (PubMed, Europe PMC, Semantic Scholar, OpenAlex), Docker + Traefik deployment, and JWT authentication.

EDUCATION

Systems Analysis and Development — FICR — Faculdade Imaculada Conceição do Recife | Feb/2024 — Jul/2026

Bachelor's in Computer Science (incomplete) — UFRPE — Universidade Federal Rural de Pernambuco | 2019 — 2022

LANGUAGES

Portuguese: Native | **English:** C2 Proficient — EFSET Certified | **Spanish:** Basic

CERTIFICATIONS

English C2 Proficient (EFSET) | ONE Tech Foundation G8 — Data Science + AI (Oracle/Alura) | SQL Advanced (Kaggle) | Python (Kaggle)

In progress: Microsoft DP-900 — Azure Data Fundamentals